

COSC 4400 / 5400

Compiler Construction

Lab Practical Exam #1

Extend the demonstration identifier Recognizer from lecture to recognize a new “NUM” token for simple integers. The regular expression for NUM tokens is:

$$0 \mid [1-9]([0-9])^*$$

A zero by itself is NUM(0). Digits 1 through 9, followed by any number of digits 0 through 9, are also NUM tokens. A starting zero followed immediately by another digit or letter is an ERROR_TOKEN. Any NUM token followed immediately by a letter is also an ERROR_TOKEN.

Once you have executed the “lpstart” script, run “make” to compile your Recognizer, “java Recognizer” to execute your Recognizer, and “make test” to diff your Recognizer against the expected result. Execute “lpturnin” to submit your code.

Submit early and often; no late submissions will be accepted.